

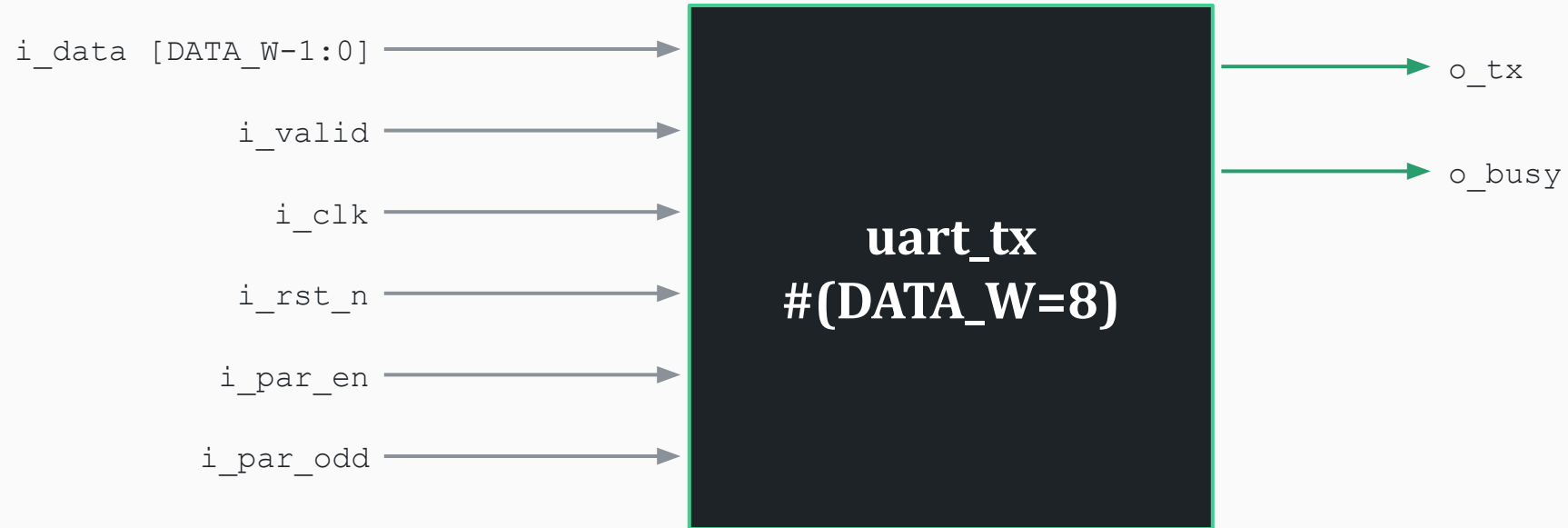
UART TX/RX — Capstone Specification

From a working transmitter to a self-checking loopback link

Roadmap

- | | | |
|---|-----------------------------------|------|
| 1 | uart_tx — recap | HAVE |
| 2 | uart_rx — new | NEW |
| 3 | Frame format — fixed contract | NEW |
| 4 | What you deliver, how it's graded | NEW |

uart_tx — Port Diagram

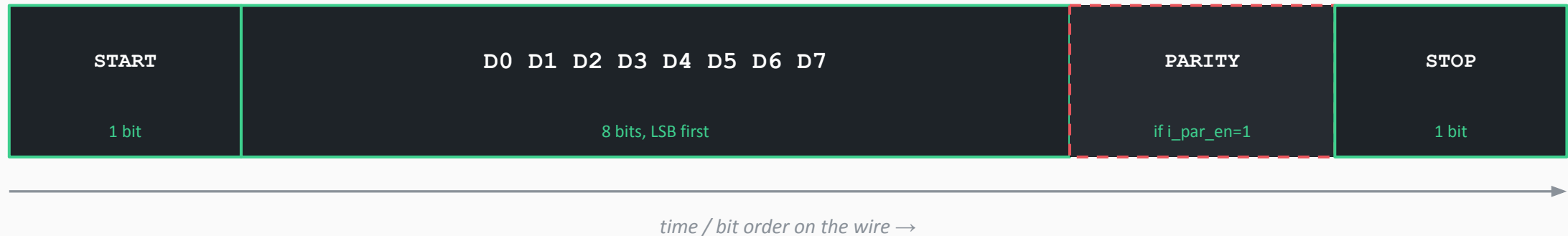


Module name and ports are the contract — copy exactly, per §4 of the student spec.

uart_tx — Required Behaviour

- Out of reset: `o_tx = 1`, `o_busy = 0`
- A byte is accepted only when `i_valid = 1` AND `o_busy = 0` — same cycle, both true
- `i_valid` while `o_busy = 1` is ignored — no queuing
- After acceptance, `o_tx` carries the frame one bit per cycle, then returns to idle-high
- `o_busy = 1` for exactly the frame duration (10 or 11 cycles), 0 otherwise
- Changing `i_data` after the accept cycle must not affect the frame already in flight

Frame Format — Fixed Contract (§2)

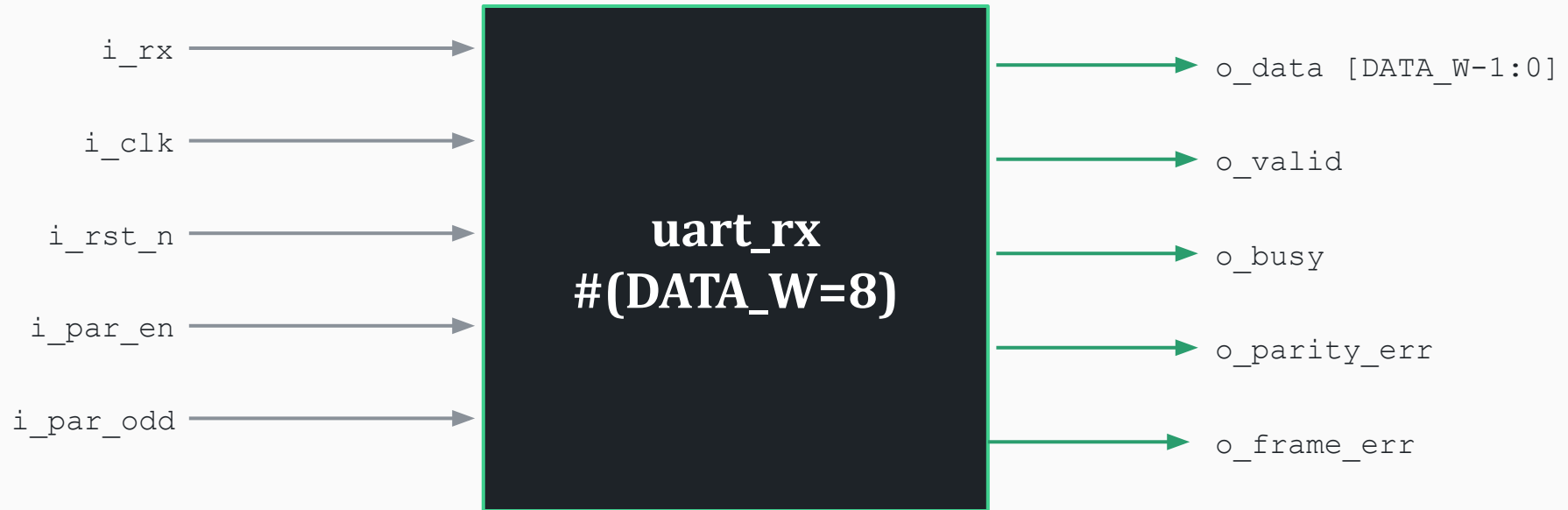


- 1 system clock cycle = 1 bit period — no baud generator, no oversampling
- 10 cycles/frame with `i_par_en=0`, 11 cycles/frame with `i_par_en=1`
- $\text{parity_bit} = (\text{^data}) \wedge \text{i_par_odd}$ — present only when `i_par_en=1`, never padded

**TX drives frames.
RX has to find them.**

NEW

uart_rx — Port Diagram



New — mirrors uart_tx's structure where it applies. Ports per §5, copied exactly.

uart_rx — Required Behaviour

- Out of reset: o_busy = 0, outputs low
- Detects the start bit only while idle — a 0 inside the data byte is never a new start
- After a full frame, o_valid pulses for exactly 1 cycle with o_data holding the received byte
- Flag-and-deliver: even on error, o_valid still pulses and o_data still carries the byte
- o_frame_err pulses (same cycle as o_valid) when the stop-bit position isn't 1
- o_parity_err pulses when i_par_en=1 and received parity disagrees — always 0 when i_par_en=0

DISCUSSION

The line idles high. A data byte can contain a 0.

How does RX know a 0 in the middle of a frame isn't a new start bit?

Reason it out before the next slide.

Start-Bit Detection

- RX watches i_rx for a 1→0 edge only while idle (o_busy low)
- Once detected: o_busy asserts, RX counts cycles per the frame format — it stops watching for edges entirely
- A 0 inside a data byte can't restart anything — RX isn't in edge-detection mode during that window
- This is structural, not an explicit "ignore new input while busy" check the way TX's i_valid rule is

```
module edge_detector (  
    input clk , n_rst, A,  
    output pos_edge_A, neg_edge_A, edge_A);  
  
endmodule
```

Why RX Needs Its Own `i_par_en` / `i_par_odd`

- Parity config is a side-channel input on TX, not part of the wire frame
- RX can't discover parity presence or polarity from the line alone
- Resolution: RX gets the same `i_par_en` / `i_par_odd` inputs, kept in agreement with TX (the grading testbench drives both ends the same way)
- Mirrors how real UART links are configured ahead of time on both ends
- This also makes `o_frame_err` well-defined — RX knows exactly which cycle the stop bit should land on

Signal Summary — uart_tx vs uart_rx

	TX	RX
Clock / Reset	i_clk, i_rst_n (async, active-low)	i_clk, i_rst_n (async, active-low)
Data	i_data [DATA_W-1:0] (in)	o_data [DATA_W-1:0] (out)
Valid strobe	i_valid (in, 1-cyc pulse)	o_valid (out, 1-cyc pulse)
Busy	o_busy (out)	o_busy (out)
Serial line	o_tx (out, idles high)	i_rx (in)
Parity config	i_par_en, i_par_odd (in)	i_par_en, i_par_odd (in, must match TX)
Error flags	– (generates only)	o_parity_err, o_frame_err (out)

What You Deliver

You Deliver

- `uart_tx.sv`
- `uart_rx.sv`
- (+ any sub-files they need)

Provided

- `uart_loopback_tb.sv`
 - your self-checking bring-up testbench

Coding Rules — Enforced

- SystemVerilog only — logic, never reg/wire
- always_ff / always_comb — never bare always
- <= in sequential blocks, = in combinational
- Async active-low reset, if (!i_rst_n) first
- FSMs: unique case with a default, two-block style
- Named port connections only, instances prefixed u_
- No magic numbers — parameterise widths and counts
- `default_nettype none / wire at top and bottom
- File header comment on every file
- Zero lint / simulator warnings

Grading — 7 Phases (Not Shown Before Submission)

#	What it does	Expectation
1	~20 random bytes, i_par_en = 0	data matches; o_parity_err stays low
2	~20 random bytes, even parity	data matches; no error flags
3	~20 random bytes, odd parity	data matches; no error flags
4	bit error injected on parity-bit cycle	o_parity_err on that o_valid; byte still delivered
5	bit error injected on stop-bit cycle	o_frame_err on that o_valid; byte still delivered
6	back-to-back frames, new i_valid the cycle o_busy clears	every byte received
7	i_valid pulsed while o_busy is high	load ignored — no extra frame

Build uart_rx → bring it up with the loopback TB → submit for grading.